



IPL 2022 Capstone Project

The Indian Premier League (IPL) is a professional T20 cricket league in India, featuring franchises representing cities. This project explores IPL 2022 match-level data to derive meaningful insights and understand match outcomes, player performances, and team dynamics.

These are some of the important columns that we'll focus on for meaningful insights in this project.

column names: Variable Type

- date : string
- venue : string
- stage : string
- team1 : string
- team2 : string
- toss_winner : string
- toss_decision : string
- first_ings_score : integer
- second_ings_score : integer
- match_winner : string
- won_by : string
- margin : integer
- player_of_the_match : string
- top_scorer : string
- highscore : integer
- best_bowling : string
- best_bowling_fgure : string
gure : string

Loading the Libraries and Dataset

```
In [1]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")

df = pd.read_csv( 'IPL.csv' )

df.head()
```

Out[1]:

	match_id	date	venue	team1	team2	stage	toss_winner	tos
0	1	March 26,2022	Wankhede Stadium, Mumbai	Chennai	Kolkata	Group	Kolkata	
1	2	March 27,2022	Brabourne Stadium, Mumbai	Delhi	Mumbai	Group	Delhi	
2	3	March 27,2022	Dr DY Patil Sports Academy, Mumbai	Banglore	Punjab	Group	Punjab	
3	4	March 28,2022	Wankhede Stadium, Mumbai	Gujarat	Lucknow	Group	Gujarat	
4	5	March 29,2022	Maharashtra Cricket Association Stadium,Pune	Hyderabad	Rajasthan	Group	Hyderabad	

Basic Information

In [2]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 74 entries, 0 to 73
Data columns (total 20 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   match_id                             74 non-null     int64
1   date                                 74 non-null     object
2   venue                                74 non-null     object
3   team1                                74 non-null     object
4   team2                                74 non-null     object
5   stage                                74 non-null     object
6   toss_winner                           74 non-null     object
7   toss_decision                         74 non-null     object
8   first_ings_score                      74 non-null     int64
9   first_ings_wkts                       74 non-null     int64
10  second_ings_score                     74 non-null     int64
11  second_ings_wkts                      74 non-null     int64
12  match_winner                           74 non-null     object
13  won_by                                 74 non-null     object
14  margin                                 74 non-null     int64
15  player_of_the_match                   74 non-null     object
16  top_scorer                            74 non-null     object
17  highscore                             74 non-null     int64
18  best_bowling                          74 non-null     object
19  best_bowling_figure                   74 non-null     object
dtypes: int64(7), object(13)
memory usage: 11.7+ KB
```

In []:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 74 entries, 0 to 73
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   match_id              74 non-null    int64
1   date                  74 non-null    object
2   venue                 74 non-null    object
3   team1                 74 non-null    object
4   team2                 74 non-null    object
5   stage                 74 non-null    object
6   toss_winner           74 non-null    object
7   toss_decision         74 non-null    object
8   first_ings_score      74 non-null    int64
9   first_ings_wkts       74 non-null    int64
10  second_ings_score     74 non-null    int64
11  second_ings_wkts      74 non-null    int64
12  match_winner          74 non-null    object
13  won_by                74 non-null    object
14  margin                74 non-null    int64
15  player_of_the_match   74 non-null    object
16  top_scorer            74 non-null    object
17  highscore             74 non-null    int64
18  best_bowling          74 non-null    object
19  best_bowling_figure   74 non-null    object
dtypes: int64(7), object(13)
memory usage: 11.7+ KB
```

```
In [5]: s=df.shape
        print(f"your rows are {s[0]} and cols are {s[1]}")
```

your rows are 74 and cols are 20

Check the size of rows and columns of the dataset

In []:

your rows are 74 and your columns are 20

Now let's see how many columns have null values in total.

```
In [6]: df.isna().sum()
```

```
Out[6]: match_id      0
        date          0
        venue         0
        team1         0
        team2         0
        stage         0
        toss_winner    0
        toss_decision  0
        first_ings_score 0
        first_ings_wkts 0
        second_ings_score 0
        second_ings_wkts 0
        match_winner   0
        won_by         0
        margin         0
        player_of_the_match 0
        top_scorer     0
        highscore      0
        best_bowling   0
        best_bowling_figure 0
        dtype: int64
```

```
In [ ]:
```

```
Out[ ]: match_id      0
        date          0
        venue         0
        team1         0
        team2         0
        stage         0
        toss_winner    0
        toss_decision  0
        first_ings_score 0
        first_ings_wkts 0
        second_ings_score 0
        second_ings_wkts 0
        match_winner   0
        won_by         0
        margin         0
        player_of_the_match 0
        top_scorer     0
        highscore      0
        best_bowling   0
        best_bowling_figure 0
        dtype: int64
```

Now, Here comes some Basic Questions

1. Which team won the most matches?

```
In [9]: # df.head()
        mw=df.match_winner.value_counts()
```

```
mw
```

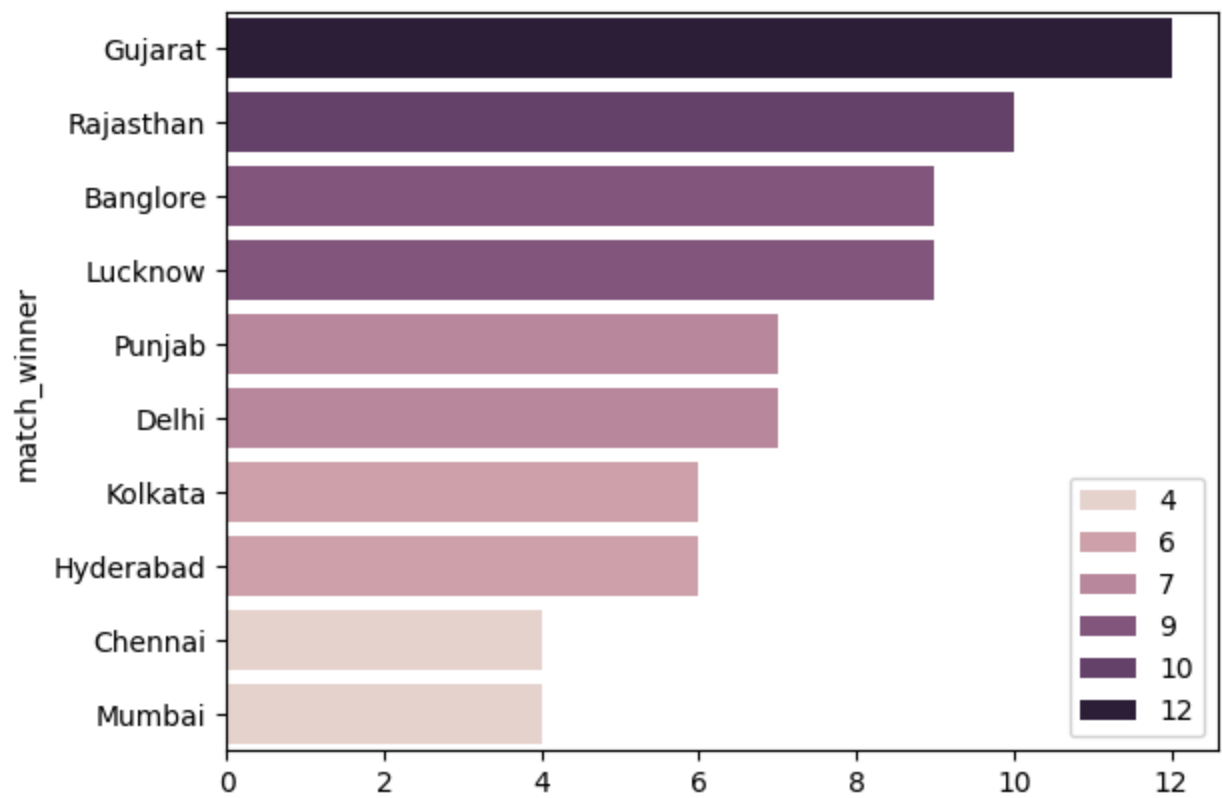
```
Out[9]: match_winner
Gujarat      12
Rajasthan    10
Banglore      9
Lucknow       9
Punjab        7
Delhi         7
Kolkata       6
Hyderabad     6
Chennai       4
Mumbai        4
Name: count, dtype: int64
```

```
In [ ]:
```

```
Out[ ]: match_winner
Gujarat      12
Rajasthan    10
Banglore      9
Lucknow       9
Delhi         7
Punjab        7
Kolkata       6
Hyderabad     6
Chennai       4
Mumbai        4
Name: count, dtype: int64
```

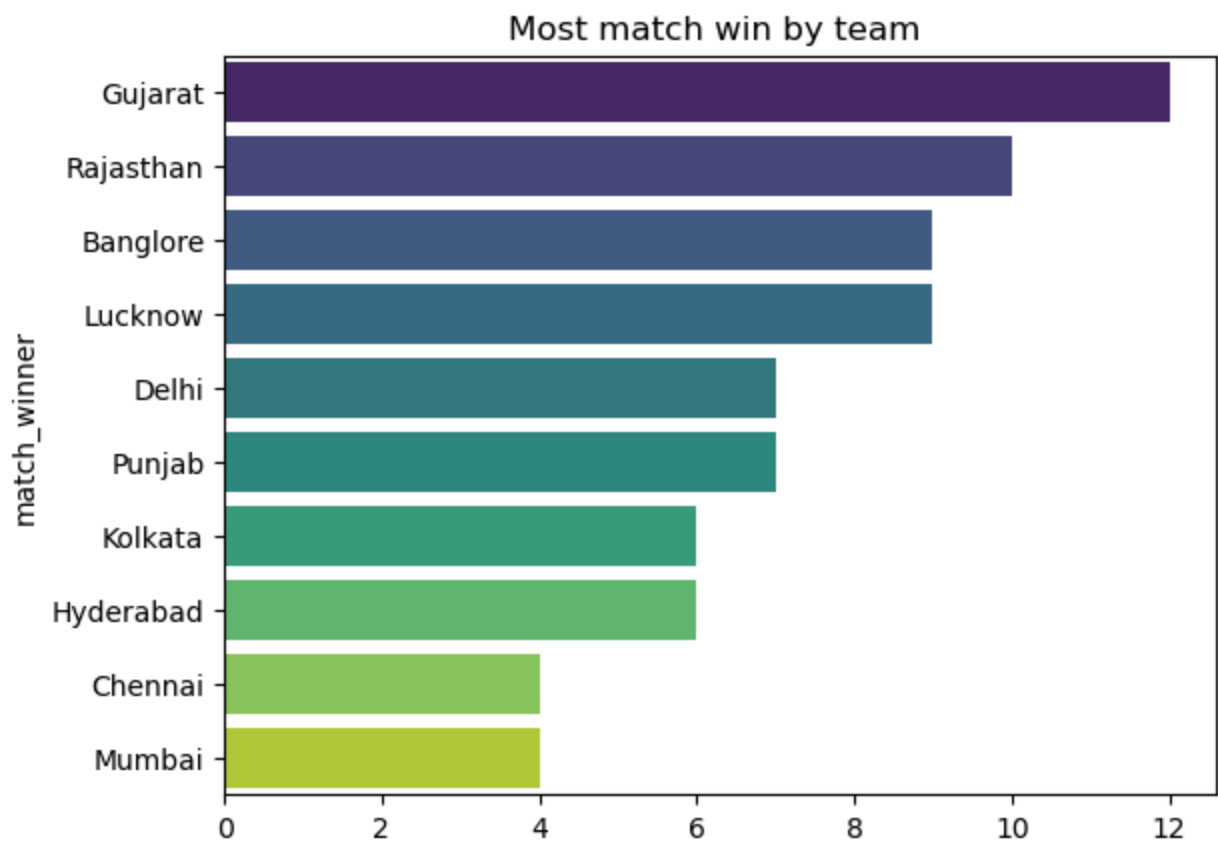
```
In [11]: sns.barplot(x= mw.values, y= mw.index, hue=mw.values)
```

```
Out[11]: <Axes: ylabel='match_winner'>
```



In []:

Out[]: Text(0.5, 1.0, 'Most match win by team ')



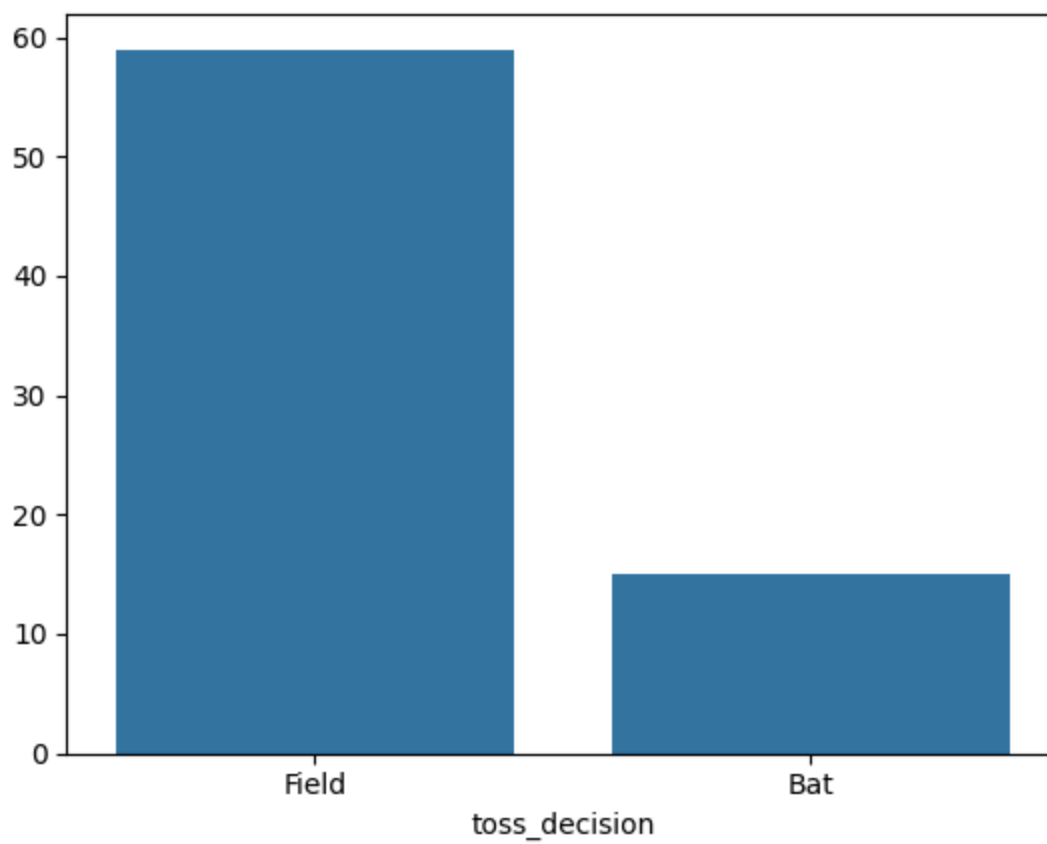
2. Toss Decision Trends

```
In [14]: Td=df.toss_decision.value_counts()  
Td
```

```
Out[14]: toss_decision  
Field    59  
Bat      15  
Name: count, dtype: int64
```

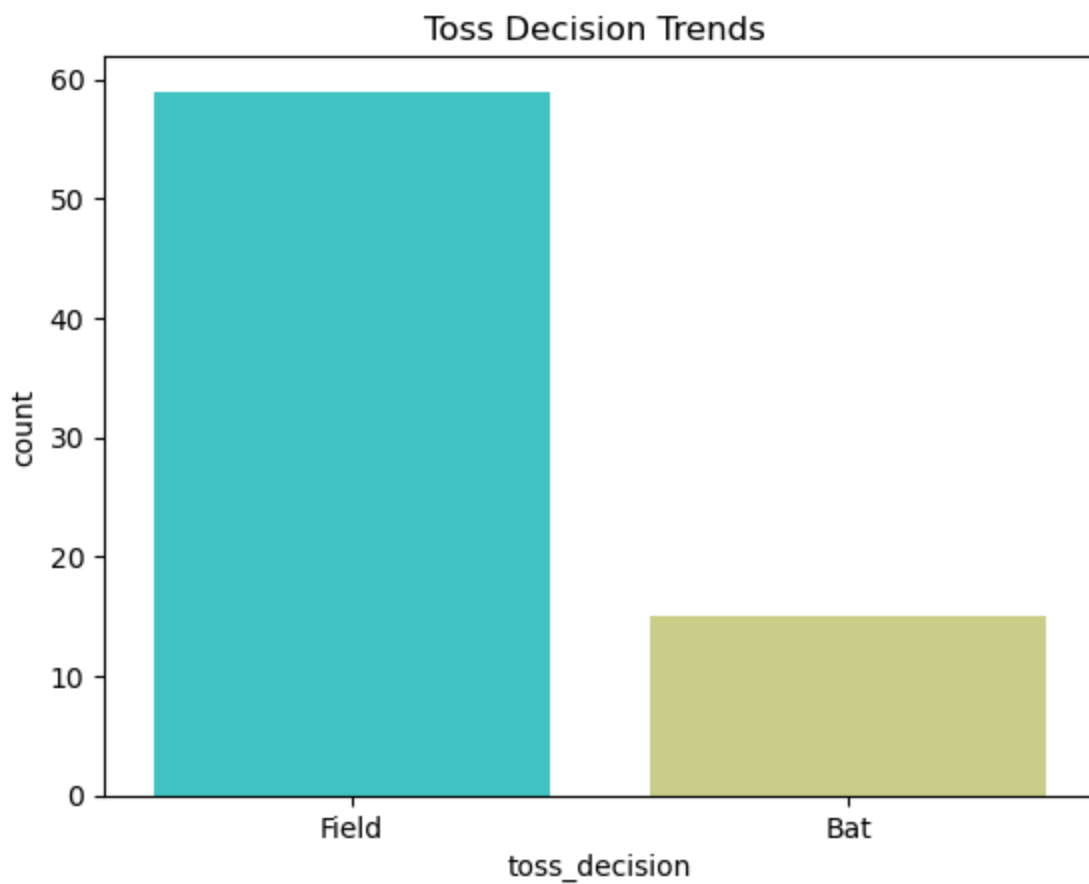
```
In [15]: sns.barplot(x=Td.index, y= Td.values)
```

```
Out[15]: <Axes: xlabel='toss_decision'>
```



In []:

Out[]: Text(0.5, 1.0, 'Toss Decision Trends ')



In []:

In []:

3. Toss Winner vs Match Winner

```
In [23]: # df.head()
t=len(df[df.toss_winner== df.match_winner])
T= len(df)
round((t/T)*100,2)
```

Out[23]: 48.65

In []:

Out[]: 48.65

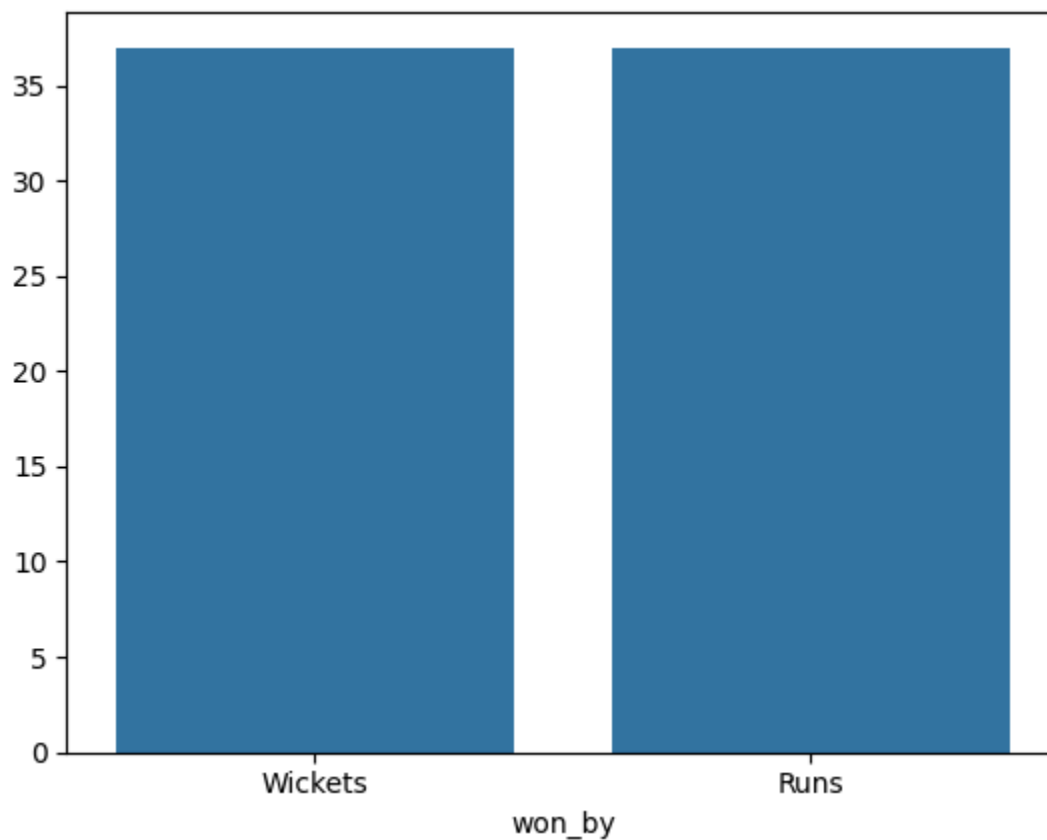
4. How do teams win? (Runs vs Wickets)

```
In [26]: df.head()
Wb=df.won_by.value_counts()
Wb
```

```
Out[26]: won_by
Wickets    37
Runs       37
Name: count, dtype: int64
```

```
In [27]: sns.barplot(x=Wb.index, y=Wb.values)
```

```
Out[27]: <Axes: xlabel='won_by'>
```



```
In [ ]:
```

Key Player Performances

1 Most "Player of the Match" Awards

```
In [31]: P=df.player_of_the_match.value_counts().head(10)
P
```

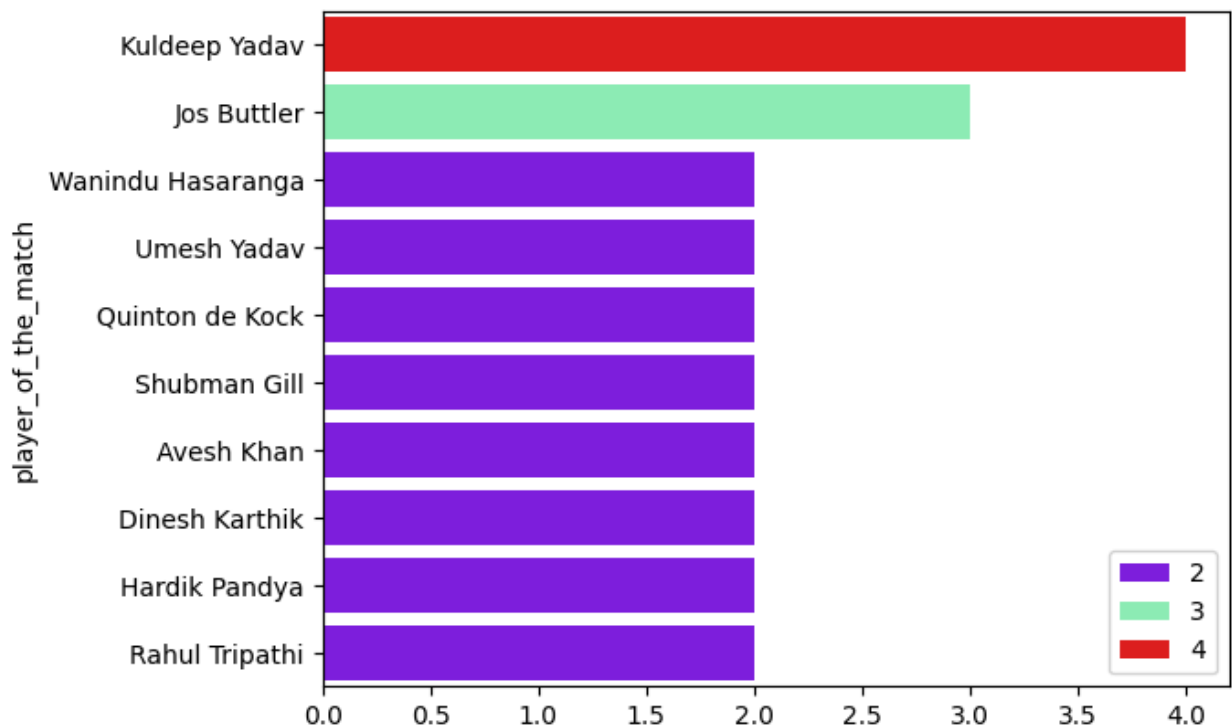
```
Out[31]: player_of_the_match
Kuldeep Yadav      4
Jos Buttler        3
Wanindu Hasaranga  2
Umesh Yadav        2
Quinton de Kock    2
Shubman Gill       2
Avesh Khan         2
Dinesh Karthik     2
Hardik Pandya      2
Rahul Tripathi     2
Name: count, dtype: int64
```

```
In [ ]:
```

```
Out[ ]: player_of_the_match
Kuldeep Yadav      4
Jos Buttler        3
Umesh Yadav        2
Quinton de Kock    2
David Miller       2
Umaran Malik       2
K L Rahul          2
Rahul Tripathi     2
Hardik Pandya      2
Jasprit Bumrah     2
Name: count, dtype: int64
```

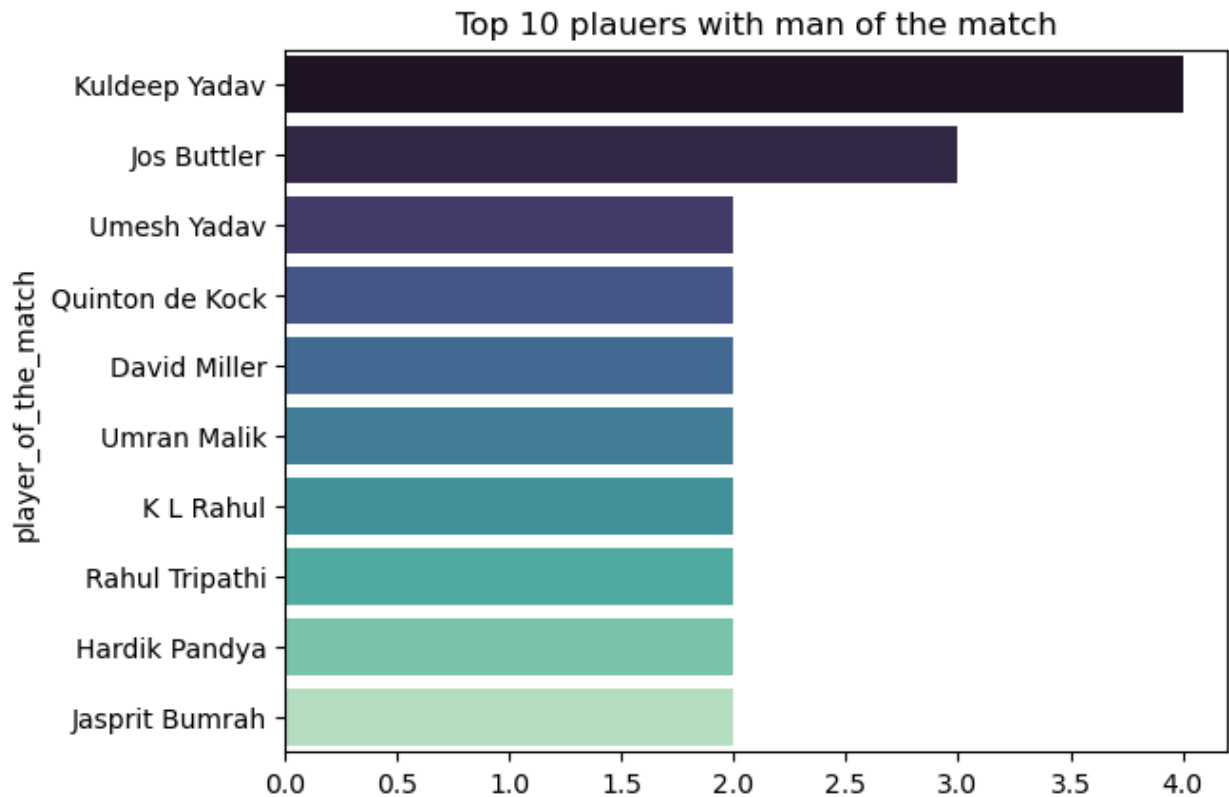
```
In [35]: sns.barplot(x= P.values, y= P.index, hue=P.values, palette="rainbow" )
```

```
Out[35]: <Axes: ylabel='player_of_the_match'>
```



```
In [ ]:
```

```
Out[ ]: Text(0.5, 1.0, 'Top 10 plauers with man of the match')
```



2 Top Scorers

```
In [41]: df.groupby("top_scorer")["highscore"].sum().sort_values(ascending=False).head(
```

```
Out[41]: top_scorer
Jos Buttler      651
Quinton de Kock  377
Name: highscore, dtype: int64
```

```
In [37]: df.top_scorer.value_counts().head(2)
```

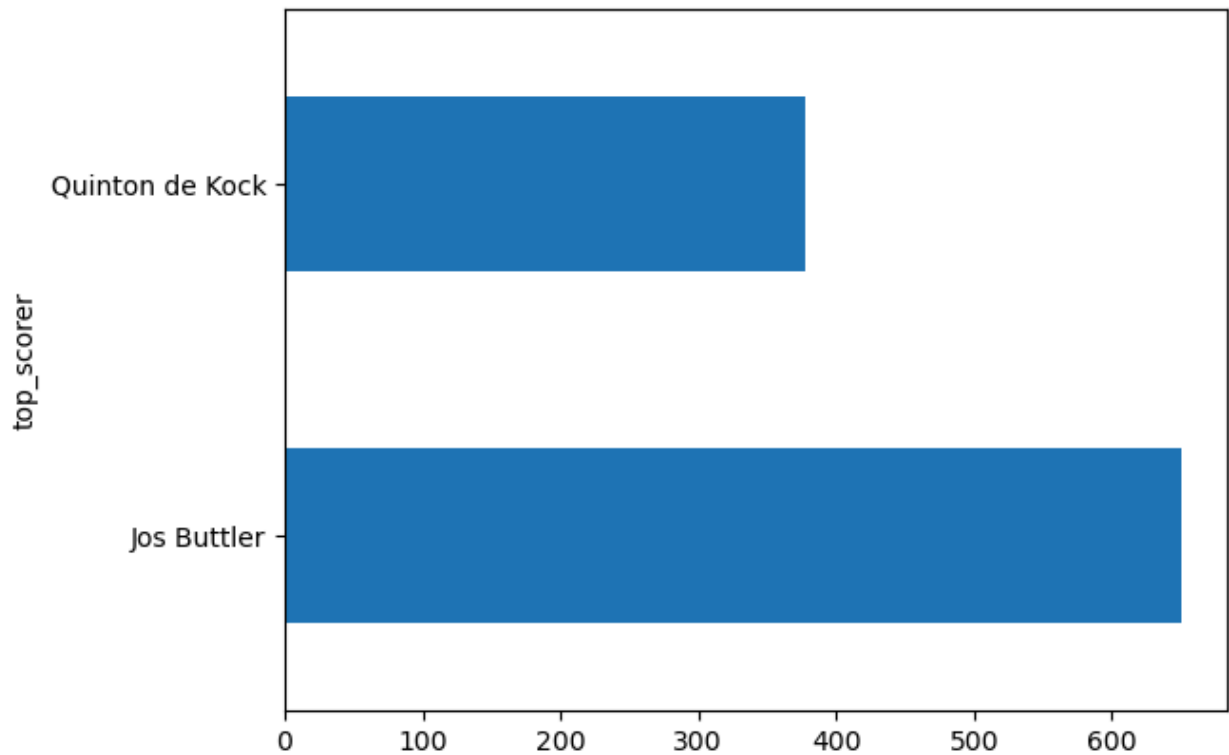
```
Out[37]: top_scorer
Jos Buttler      7
Quinton de Kock  5
Name: count, dtype: int64
```

```
In [ ]:
```

```
Out[ ]: top_scorer
Jos Buttler      651
Quinton de Kock  377
Name: highscore, dtype: int64
```

```
In [ ]:
```

Out[]: <Axes: ylabel='top_scorer'>



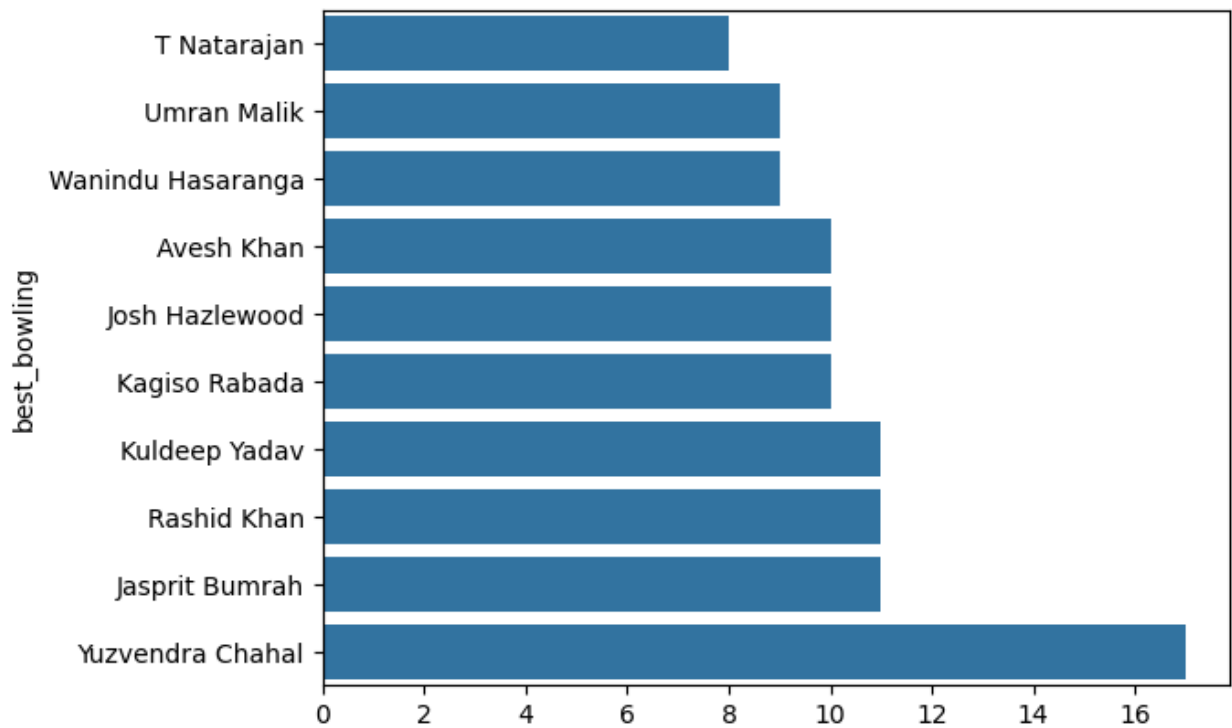
10 Best Bowling Figures

```
In [42]: df["bowling"] = df.best_bowling_figure.apply(lambda x: x.split("--")[0]).astype(int)
```

```
In [49]: # 3--20-----> [3,20]
df.head()
B=df.groupby("best_bowling")["bowling"].sum().sort_values(ascending=False).head(10)
B=B.sort_values()
```

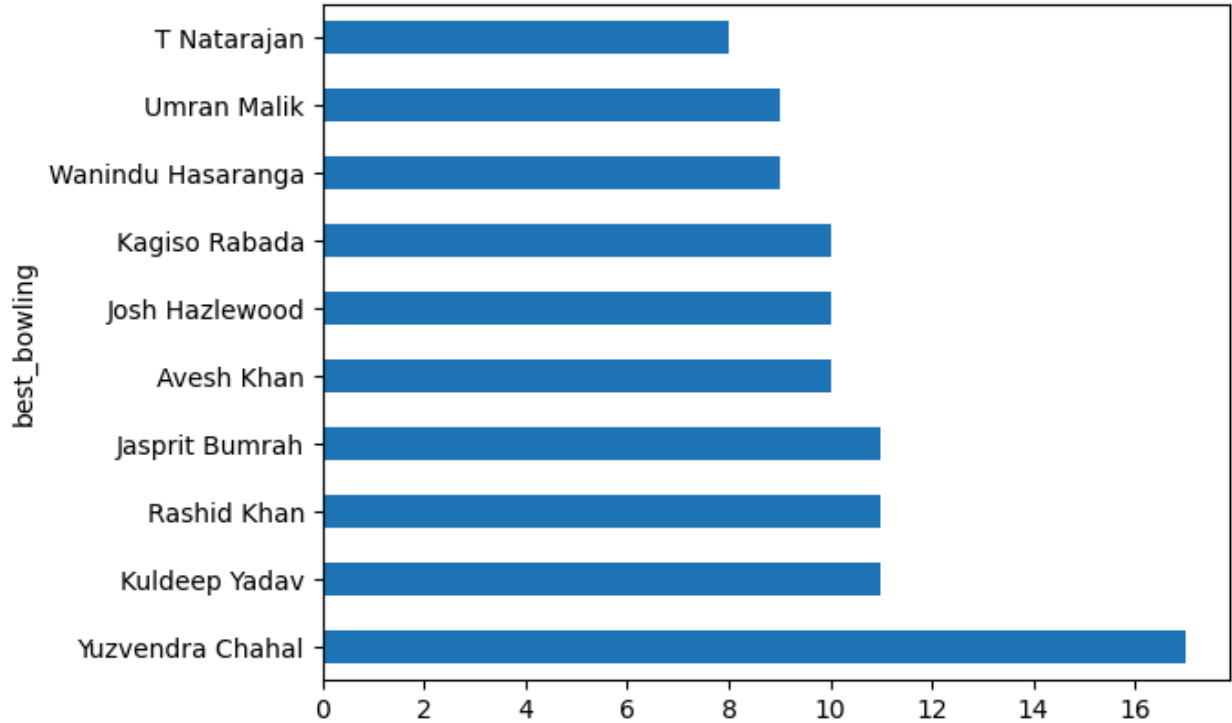
```
In [50]: sns.barplot(x=B.values, y= B.index)
```

Out[50]: <Axes: ylabel='best_bowling'>



In []:

Out[]: <Axes: ylabel='best_bowling'>



Venue Analysis

Most Matches Played by Venue

```
In [52]: df.venue.value_counts()
```

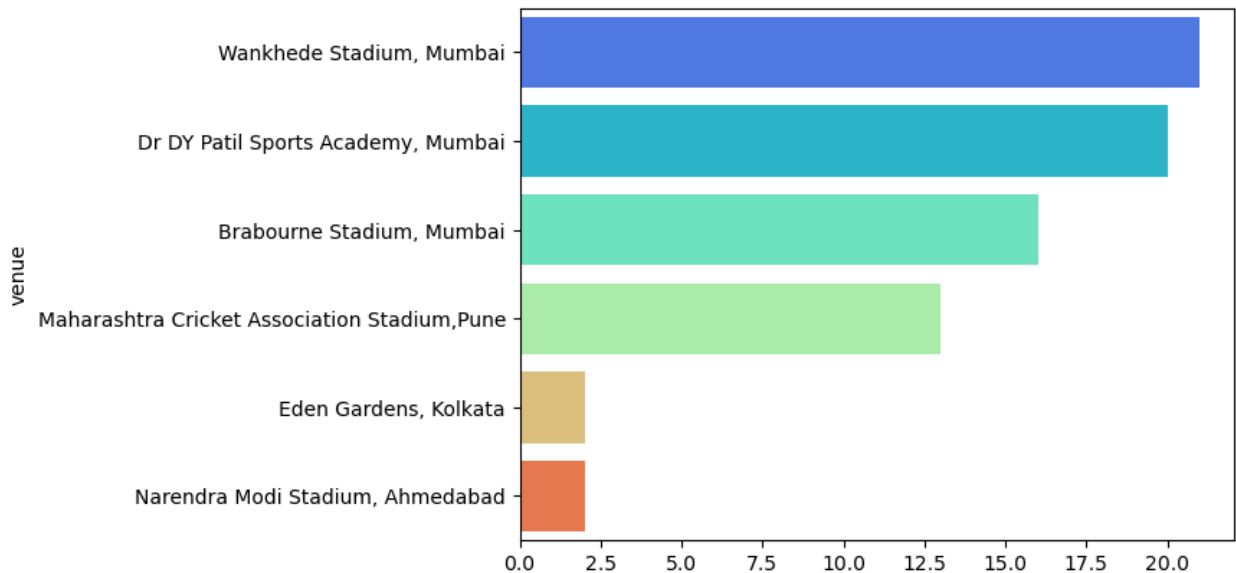
```
Out[52]: venue
Wankhede Stadium, Mumbai      21
Dr DY Patil Sports Academy, Mumbai  20
Brabourne Stadium, Mumbai     16
Maharashtra Cricket Association Stadium,Pune  13
Eden Gardens, Kolkata         2
Narendra Modi Stadium, Ahmedabad  2
Name: count, dtype: int64
```

```
In [ ]:
```

```
Out[ ]: venue
Wankhede Stadium, Mumbai      21
Dr DY Patil Sports Academy, Mumbai  20
Brabourne Stadium, Mumbai     16
Maharashtra Cricket Association Stadium,Pune  13
Eden Gardens, Kolkata         2
Narendra Modi Stadium, Ahmedabad  2
Name: count, dtype: int64
```

```
In [ ]:
```

```
Out[ ]: <Axes: ylabel='venue'>
```



Custom Questions & Insights

Q1: Who won the highest margin by runs?

```
In [58]: df.sort_values(by="margin", ascending=False)[["match_winner", "margin"]].head()
```

```
Out[58]:
```

	match_winner	margin
54	Chennai	91

```
In [ ]:
```

```
Out[ ]:
```

	match_winner	margin
54	Chennai	91

Q2: Which player had the highest individual score?

```
In [62]: df.head()  
# df.highscore.max()  
df.sort_values(by= "highscore", ascending=False)[["top_scorer", "highscore"]].
```

```
Out[62]:
```

	top_scorer	highscore
65	Quinton de Kock	140

```
In [ ]:
```

```
Out[ ]:
```

	top_scorer	highscore
65	Quinton de Kock	140

Q3: Which bowler had the best bowling figures?

```
In [71]: df.head()  
# df.groupby("best_bowling")["bowling"].sum()  
df.sort_values(ascending=False, by="bowling")[["best_bowling", "best_bowling_f
```

```
Out[71]:
```

	best_bowling	best_bowling_figure
29	Yuzvendra Chahal	5--40
39	Umaran Malik	5--25
55	Jasprit Bumrah	5--10
53	Wanindu Hasaranga	5--18

```
In [ ]:
```


Out[]:

	best_bowling	best_bowling_figure
29	Yuzvendra Chahal	5--40
39	Umran Malik	5--25
53	Wanindu Hasaranga	5--18
55	Jasprit Bumrah	5--10